Additional implementation qualifications

Unpromoted experiments; delivery remains incomplete. These results qualify individual experimental changes. They do not change the production source, registered campaign, or acceptance status above.

The optimized QP caller currently squares denominator masses in binary64 before widening. An experimental change across five source paths passes the model's QP masses and squares them at QP only when backend 9 is actually selected. In the QCD process $u\bar{u} \to Zgg$, original and candidate each complete normal learning and 100 saved final points with 87 DP and 13 naturally rescued QP answers. All 800 amplitude-coefficient words and 200 accuracy words match exactly with native integral caching both disabled and enabled. Native-input observation confirms 1,118 corrected 105-bit bottom-mass squares and 72 changed mass-dependent invariants; numerator, momentum-shift and scale records remain identical. Separate recording-stub controls preserve genuine 113-bit Fortran input words. The physical real-model test does not itself supply 113-bit model masses.

A subsequent fresh MG CLI export, complete generated-source review, compilation and 100-point amplitude run reproduce the corrected candidate's coefficient, accuracy, status and observed exception states exactly. This run uses the QP integrand, native cache disabled, and normal independent stability checks. The review checks all 195 generated source/build-input bindings, 25 unchanged public routines, nine private QP routines and 116 guarded calls; every changed scientific source is recompiled. The earlier failed CLI startup and the historical generation result marked `generated_pending_source_review` remain retained. The later chain supplies source-to-caller correspondence without rewriting those outcomes. Electroweak, loop-induced, complex-mass, five-backend and independent finite-answer qualification remain open, and the patch is unapplied.

A separate Fortran profile measures the existing phase-space repair in $u\bar{u} \to Z$ with independent stability checks. Across seven four-lane cycles and 358,400 timed full-amplitude calls, the median inclusive DP repair is 6.075 µs within a clock-instrumented 89.914 µs amplitude; its nested MP repair is 5.627 µs. These intervals overlap and must not be added. Instrumented and plain lanes preserve final numerical words and observed exception states. These descriptive shared-host measurements identify a modest part of the total runtime; they demonstrate no optimization gain or formal performance acceptance.

Two standalone Fortran components support further complex-mass research. A bounded integer exact-geometry helper passes 67,920 calls across 20 floating-point-environment cohorts, explicitly refusing unprovable cases. A caller-owned snapshot module passes 1,480 observations, 3,680 API outcomes and 94,720 state-word checks under the same 20 cohorts, with checked and optimized builds agreeing. These tests qualify equality checks and snapshot transport only. Actual constructor ownership, use of the current physical point, final scalar products and the complete complex-mass caller still require numerical qualification.

The preceding five-backend timing tables, including the 2⟶4 case, and Appendix-C-style stability figures retain their original datasets and source identities. Native-DP stability and rescued-answer accuracy remain separate; 100 samples resolve fractions only down to 1%. Hua-Sheng's local paired timing run remains the formal performance TODO. None of these experiments changes a failed or unmeasured main-campaign row into a passing result.

Exact supplementary index hashes and scope are supplied in `qualification-selection.json`. The following links provide raw evidence and reproduction commands.

QP-mass stages · Fresh-CLI followup · Repair profile · Integer equality · Snapshot transport.